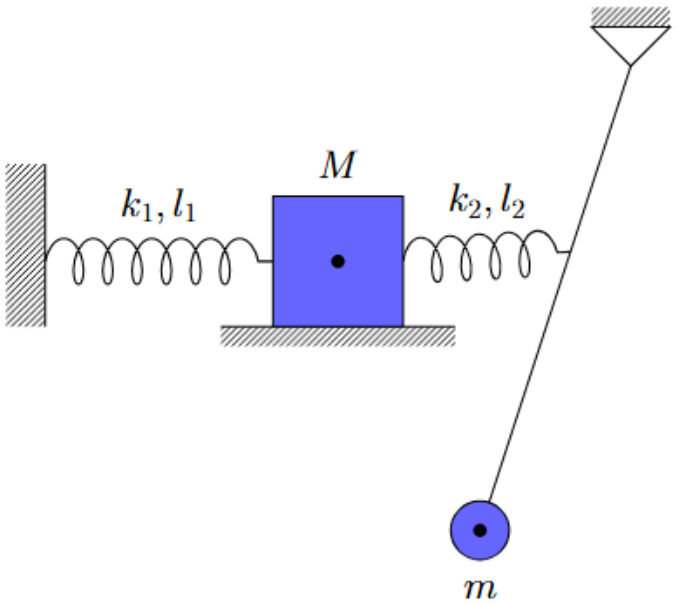
HELMO GRAMME
Ecole d'ingénieur·e·s

UE 4 - Modelling and system analysis

Report No. 1



Teachers
VIGNERON Didier
HOFFAIT Anne-Sophie
LETAWE Yannick

Authors
RASSE Simon
KALSCHEUER Tom
DELHAXHE Pierre

2026–2027 academic year

Contents

1 Introduction and system definition **3**
1.1 System Selection and Constraints 3
1.2 Definition of Variables and Frames of Reference 4

2 Kinematics **5**
2.1 Position Vectors and Constraints 5
2.2 Velocities and Accelerations 5

3 Dynamics **6**
3.1 Analysis of Mass m 6
3.2 Analysis of Mass M 7
3.3 Lagrangian Formulation 8

4 Mathematical Modeling **10**
4.1 Dimensionless form of the mathematical model 10

1 Introduction and system definition

This section is dedicated to explaining our system in general. In Section 1.1, we will present the system and verify that it meets all the required conditions. In Section 1.2, we will define all the variables used in the system.

1.1 System Selection and Constraints

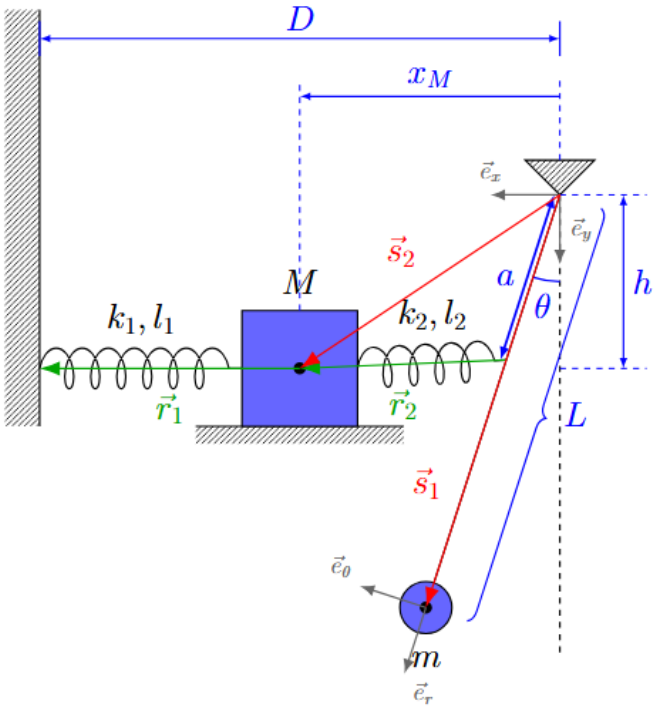


Figure 1: System diagram

The system we have chosen is shown in Figure 1. We designed this specific system because it strictly meets all the course's criteria. Our system features:

1. exactly two degrees of freedom,
2. coupled non-linear differential equations,
3. a stable equilibrium point.

Regarding the degrees of freedom, the spatial configuration is fully determined by two independent coordinates: the horizontal position x_M of mass M and the angle θ of the pendulum relative to the vertical.

It is easy to see that the system is governed by coupled differential equations because the elongation of the spring connecting the two bodies depends on both the position x of mass M and the angle θ. Consequently, the equation describing the acceleration of mass M ($\ddot{x_M}$) will contain terms involving θ. Exactly the same applies to the angular acceleration ($\ddot{\theta}$) of the pendulum.

While we will formally demonstrate the non-linearity of the differential equations once Newton's equations of motion are written down, it can already be anticipated due to the rotational motion of the pendulum, which inevitably introduces trigonometric functions.

Finally, again, we can't prove at this stage that this system has a stable equilibrium point, but instinctively, we see that the system does indeed possess a stable equilibrium point. Indeed, we know that the movement of a pendulum has a stable equilibrium point and in our system we have a pendulum but this pendulum is retained by a system with springs so we can suppose that the entire system possess a stable equilibrium point that will be slightly offset from that of a simple pendulum.

1.2 Definition of Variables and Frames of Reference

Referring to Figure 1.1, we can now define all the relevant variables and frames of reference.

The $\vec{s}$ vectors represent the positions of the masses, while the $\vec{r}$ vectors define the elongation of the springs.

We use two frames of reference: the Cartesian frame $(O, \vec{e}_x, \vec{e}_y)$ and the polar frame $(\vec{e}_r, \vec{e}_\theta)$.

2 Kinematics

This section is dedicated to the kinematics of the system. In Section 2.1, we will express all the vectors ($\vec{s}$ and $\vec{r}$) in the Cartesian and polar coordinate systems using Figure 1.1. In Section 2.2, we will derive the expressions for the acceleration of each mass.

2.1 Position Vectors and Constraints

We can now express these vectors in the Cartesian frame (Oxy) and the polar frame $(\vec{e}_r, \vec{e}_\theta)$:

$$\begin{aligned}
\vec{s}_1 &= L\vec{e}_r \\
\vec{s}_2 &= x_M\vec{e}_x + h\vec{e}_y \\
\vec{r}_2 &= \vec{s}_2 - a\vec{e}_r \\
&= (x_M - a\sin\theta)\vec{e}_x + (h - a\cos\theta)\vec{e}_y \\
&= (x_M\sin\theta + h\cos\theta - a)\vec{e}_r + (x_M\cos\theta - h\sin\theta)\vec{e}_\theta
\end{aligned}$$

$$\vec{r}_1 = D\vec{e}_x + h\vec{e}_y - \vec{s}_2 = (D - x_M)\vec{e}_x.$$

Where:

$$\begin{cases} \vec{e}_r = \sin\theta\,\vec{e}_x + \cos\theta\,\vec{e}_y \\ \vec{e}_\theta = \cos\theta\,\vec{e}_x - \sin\theta\,\vec{e}_y. \end{cases}$$

2.2 Velocities and Accelerations

To find the expressions for the acceleration of each mass, we simply have to differentiate the mathematical expression for the position of the mass twice.

Mass m:

$$\begin{aligned}
\vec{s}_1 &= L\sin\theta\,\vec{e}_x + L\cos\theta\,\vec{e}_y \\
\vec{v}_1 &= L\cos\theta\,\dot{\theta}\,\vec{e}_x - L\sin\theta\,\dot{\theta}\,\vec{e}_y \\
\vec{a}_1 &= (-L\sin\theta\,\dot{\theta}^2 + L\cos\theta\,\ddot{\theta})\vec{e}_x + (-L\cos\theta\,\dot{\theta}^2 - L\sin\theta\,\ddot{\theta})\vec{e}_y
\end{aligned} \tag{1}$$

Or alternatively, using the polar basis:

$$\begin{aligned}
\vec{s}_1 &= L\vec{e}_r \\
\vec{v}_1 &= L\dot{\theta}\,\vec{e}_\theta \\
\vec{a}_1 &= L\ddot{\theta}\,\vec{e}_\theta - L\dot{\theta}^2\,\vec{e}_r
\end{aligned} \tag{2}$$

Mass M:

$$\begin{aligned}
\vec{s}_2 &= x_M\vec{e}_x + h\vec{e}_y \\
\vec{v}_2 &= \dot{x}_M\vec{e}_x \\
\vec{a}_2 &= \ddot{x}_M\vec{e}_x
\end{aligned} \tag{3}$$

3 Dynamics

This section is dedicated to establishing the dynamic equilibrium of the system using Newton's second law and verifying the equations using the Langrangian formulation. To do so, we isolate each mass and establish its Free Body Diagram.

3.1 Analysis of Mass m

We first isolate the pendulum mass m. The Free Body Diagram is shown in Figure 2.

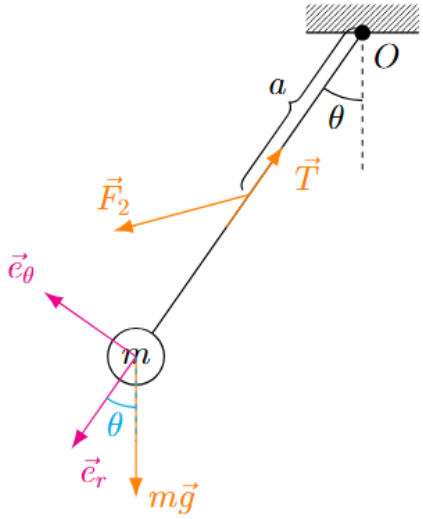


Figure 2: Free Body Diagram of mass m

The forces acting on mass m are:

- The gravitational force: $\vec{P}_m = mg\,\vec{e}_y$
- The tension from the rod: $\vec{T} = -T\,\vec{e}_r$
- The force exerted by spring k_1: $\vec{F}_2 = k_2(\|\vec{r_2}\| - l_2)\dfrac{\vec{r_2}}{\|\vec{r_2}\|}$

Since the mass m is constrained to rotate around the hinge O, it is most appropriate to apply the angular momentum principle around point O, projected along the $\vec{e}_z$ axis:

$$\sum M_O = I\ddot{\theta} = mL^2\ddot{\theta}. \tag{4}$$

By computing the moments of the applied forces $\vec{P}_m$, $\vec{T}$, and $\vec{F}_2$ around O, we obtain the scalar differential equation of motion for the pendulum:

$$\boxed{k_2\left(1 - \frac{l_2}{\|\vec{r_2}\|}\right)(x_M\cos\theta - h\sin\theta)a - mg\sin\theta L = mL^2\ddot{\theta}.} \tag{5}$$

Finally, we apply Newton's law to mass m projected along the rod axis $\vec{e}_r$ witch gives

$$T = mg\cos\theta + mL\dot{\theta}^2 + k_2\left(1 - \frac{l_2}{\|\vec{r}_2\|}\right)(x_M \sin\theta + h\cos\theta - a). \tag{6}$$

3.2 Analysis of Mass M

Next, we isolate the mass M sliding on the horizontal rail. The Free Body Diagram is presented in Figure 3.

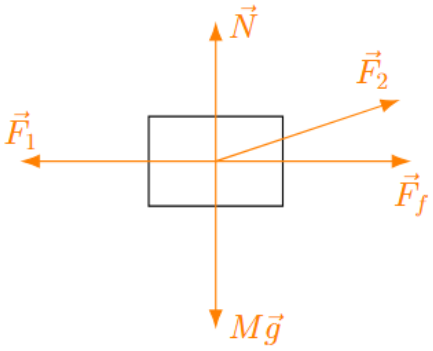


Figure 3: Free Body Diagram of mass M

The forces acting on mass M are:

- The gravitational force: $\vec{P}_M = Mg\,\vec{e}_y$
- The normal reaction from the rail: $\vec{N} = -N\,\vec{e}_y$
- The force exerted by spring k_2: $\vec{F}_2 = -k_2(\|\vec{r}_2\| - l_2)\dfrac{\vec{r}_2}{\|\vec{r}_2\|}$
- The force exerted by spring k_1: $\vec{F}_1 = k_1(\|\vec{r}_1\| - l_1)\vec{e_x}$
- The frictional forces : $\vec{F}_f = -\beta v_M \vec{e_x}$.

Applying Newton's second law to mass M yields:

$$\sum \vec{F} = M\vec{a} \tag{7}$$

Substituting the identified forces into the equation, we obtain this equation:

$$\vec{F}_1 + \vec{F}_2 + \vec{N} + \vec{P}_M + \vec{F}_f = M\ddot{x}_M\vec{e}_x \tag{8}$$

By projecting this vector equation onto the Cartesian axes $(O, \vec{e}_x, \vec{e}_y)$, we obtain the scalar equations for the mass M.

Along the $\vec{e}_x$ axis, we obtain the differential equation of motion:

$$\boxed{M\ddot{x}_M = k_1(\|\vec{r}_1\| - l_1) - k_2\left(1 - \frac{l_2}{\|\vec{r}_2\|}\right)(x_M - a\sin\theta) - \beta\dot{x}_M} \tag{9}$$

Along the $\vec{e}_y$ axis, the acceleration is zero, which allows us to determine the normal reaction force:

$$N = Mg - k_2 \left(1 - \frac{l_2}{\|\vec{r}_2\|}\right)(h - a\cos\theta) \tag{10}$$

3.3 Lagrangian Formulation

To verify the conservative part of the equations of motion derived from Newton's second law, we will re-establish the system's dynamics using the standard Lagrangian formulation. The Lagrangian $\mathcal{L}$ of the system is defined as the difference between the total kinetic energy E_k and the total potential energy E_p:

$$\mathcal{L} = E_k - E_P \tag{11}$$

Based on the kinematics established in Section 2, the total kinetic energy of the system is the sum of the translational kinetic energy of mass M and the rotational kinetic energy of mass m:

$$E_k = \frac{1}{2}M\dot{x}_M^2 + \frac{1}{2}mL^2\dot{\theta}^2 \tag{12}$$

The total potential energy includes the gravitational potential energy of both masses and the elastic potential energy stored in the two springs. Let r_{mid} be the length of the central spring k_2, such that $r_{mid} = \sqrt{(x_M - a\sin\theta)^2 + (h - a\cos\theta)^2}$. The potential energy is given by:

$$E_p = -Mgh - mgL\cos\theta + \frac{1}{2}k_1(D - x_M - l_1)^2 + \frac{1}{2}k_2(r_{mid} - l_2)^2 \tag{13}$$

The dynamics of the system are governed by the Euler-Lagrange equations:

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_i}\right) - \frac{\partial\mathcal{L}}{\partial q_i} = 0 \tag{14}$$

where $q_i \in \{x_M, \theta\}$.

Equation of motion for x_M

Applying the Euler-Lagrange equation for the coordinate x_M, we compute the required partial derivatives:

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}_M}\right) = M\ddot{x}_M$$

$$\frac{\partial\mathcal{L}}{\partial x_M} = -\frac{\partial E_p}{\partial x_M} = k_1(D - x_M - l_1) - k_2(r_{mid} - l_2)\frac{x_M - a\sin\theta}{r_{mid}}$$

Substituting these into the Euler-Lagrange equation yields:

$$M\ddot{x}_M - k_1(D - x_M - l_1) + k_2\left(1 - \frac{l_2}{r_{mid}}\right)(x_M - a\sin\theta) = 0 \tag{15}$$

Rearranging the terms, we find the differential equation (without the non-conservative friction term):

$$\boxed{M\ddot{x}_M = k_1(D - x_M - l_1) - k_2\left(1 - \frac{l_2}{r_{mid}}\right)(x_M - a\sin\theta)} \tag{16}$$

Equation of motion for θ

Similarly, applying the Euler-Lagrange equation for the angular coordinate θ:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = mL^2\ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\frac{\partial E_p}{\partial \theta} = -mgL\sin\theta - k_2(r_{mid} - l_2)\frac{a(h\sin\theta - x_M\cos\theta)}{r_{mid}}$$

Substituting these derivatives gives:

$$mL^2\ddot{\theta} + mgL\sin\theta + k_2 a\left(1 - \frac{l_2}{r_{mid}}\right)(h\sin\theta - x_M\cos\theta) = 0 \tag{17}$$

Which can be rearranged to perfectly match the Newtonian rotational equation:

$$\boxed{mL^2\ddot{\theta} = k_2 a\left(1 - \frac{l_2}{r_{mid}}\right)(x_M\cos\theta - h\sin\theta) - mgL\sin\theta} \tag{18}$$

4 Mathematical Modeling

In this section, we nondimensionalize the mathematical model governing the two-degree-of-freedom system. Using the Vaschy-Buckingham π theorem, we define the relevant dimensionless parameters to reformulate the differential equations. This reduces the system's parametric complexity and yields a generalized form optimized for numerical resolution.

4.1 Dimensionless form of the mathematical model

Before attempting to make our two equations dimensionless, we determined the number of variables involved and the number of fundamental units they represent. According to the Vaschy-Buckingham theorem, the difference between these two quantities dictates the number of independent dimensionless parameters we can construct.

Given that we have fourteen variables $(t, k_1, k_2, m, M, D, h, a, l_1, l_2, L, \beta, g)$ and three fundamental units (kg, m, s), we can define exactly eleven dimensionless numbers.

To non-dimensionalize the governing equations, we began by scaling the time variable. Thus, we defined:

$$\tau = A \cdot t \quad \text{where} \quad A = \frac{1}{2\pi}\sqrt{\frac{g}{L}}. \tag{19}$$

Here, A represents the natural oscillation frequency of a simple pendulum. Multiplying this frequency by time yields a dimensionless time τ, which corresponds to the number of pendulum oscillations.

Next, to scale the position x_M and its derivatives, we introduced:

$$\psi = \frac{x_M}{D}. \tag{20}$$

This dimensionless parameter represents the normalized position of the mass relative to the distance between the pendulum's origin and the wall.

These scaling choices allow us to express the first and second time derivatives of our variables with respect to the dimensionless time τ. Consequently, we obtain:

$$\dot{x}_M = D \cdot \frac{1}{2\pi}\sqrt{\frac{g}{L}}\frac{d\psi}{d\tau}, \tag{21}$$

$$\ddot{x}_M = D \cdot \frac{1}{4\pi^2}\frac{g}{L}\frac{d^2\psi}{d\tau^2}, \tag{22}$$

$$\dot{\theta} = \frac{1}{2\pi}\sqrt{\frac{g}{L}}\frac{d\theta}{d\tau}, \tag{23}$$

$$\ddot{\theta} = \frac{1}{4\pi^2}\frac{g}{L}\frac{d^2\theta}{d\tau^2}. \tag{24}$$

Substituting these expressions into our initial equations of motion yields:

$$\frac{MD}{4\pi^2}\cdot\frac{g}{L}\cdot\frac{d^2\psi}{d\tau^2} = k_1(D-\psi D-l_1) - \frac{\beta D}{2\pi}\cdot\sqrt{\frac{g}{L}}\cdot\frac{d\psi}{d\tau} - k_2(\psi D - a\sin\theta)\left(1-\frac{l_2}{\sqrt{(\psi D - a\sin\theta)^2+(h-a\cos\theta)^2}}\right) \tag{25}$$

and

$$\frac{mL^2}{4\pi^2}\cdot\frac{g}{L}\cdot\frac{d^2\theta}{d\tau^2} = -mgL\sin\theta + k_2 a(\psi D\cos\theta - h\sin\theta) \times\left(1-\frac{l_2}{\sqrt{(\psi D - a\sin\theta)^2+(h-a\cos\theta)^2}}\right) \tag{26}$$

From these two equations, we sought to identify meaningful dimensionless groups. Substituting these groups into the system will ultimately yield fully dimensionless equations of motion. We therefore defined the following dimensionless parameters:

$$\gamma = \frac{\sqrt{M/k_2}}{\sqrt{L/g}},\quad \mu = \frac{\sqrt{M/k_1}}{\sqrt{L/g}},\quad \xi = \frac{a}{D},\quad \lambda = \frac{l_1}{D},$$
$$\sigma = \frac{l_2}{D},\quad \alpha = \frac{h}{D},\quad \delta = \frac{k_2 l_2}{mg},\quad \nu = \frac{\beta/M}{2\pi\sqrt{g/L}},$$
$$\eta = \frac{a}{L}.$$

Physically, the parameters γ and μ represent the ratio between the natural frequencies of the springs and the pendulum. The variables $\xi, \alpha, \lambda, \sigma$, and η serve as geometric ratios that define the system's spatial configuration and spring free lengths. Furthermore, δ compares the relative magnitude of the second spring's force to gravitational effects, while ν scales the viscous damping against the pendulum's restorative force. Together with the dimensionless time τ and position ψ, this yields exactly the eleven predicted parameters. We can now write the equations of motion in their final dimensionless form:

$$\boxed{\frac{d^2\psi}{d\tau^2} = \frac{1}{\mu^2}4\pi^2(1-\psi-\lambda) - \frac{1}{\gamma^2}\epsilon(\theta,\psi)\cdot 4\pi^2(\psi-\xi\sin\theta) - \nu\frac{d\psi}{d\tau}} \tag{27}$$

and

$$\boxed{\frac{d^2\theta}{d\tau^2} = \epsilon(\theta,\psi)\cdot 4\pi^2\frac{\delta\eta}{\sigma}(\psi\cos\theta - \alpha\sin\theta) - 4\pi^2\sin\theta} \tag{28}$$

where

$$\epsilon(\theta,\psi) = 1 - \frac{\sigma}{\sqrt{\psi^2+\xi^2+\alpha^2-2\xi\psi\sin\theta-2\alpha\xi\cos\theta}}. \tag{29}$$